

INTERNSHIP REPORT

DJEN DJEN PORT - Jijel **AI Port Storage Optimization Platform** (Intelligent Port Storage Management and Optimization System)

Prepared by :
BENSABRA ALA
3rd Year Engineering Student
At National Higher School of Artificial Intelligence (ENSIA)

Internship Period :
From 06/09/2026 to 01/10/2026
(1 month internship)



Project: Intelligent Port Storage Management and Optimization System

Developed as part of a one-month internship project at Djen Djen Port, this document presents the complete technical description of the developed platform. The objective of this project was to design and implement an intelligent storage management system capable of optimizing cargo allocation, improving terminal operations, and providing decision-support tools for port management. This document serves as a technical reference for project presentation, defense, and detailed understanding of the implemented functionalities, system architecture, and development choices.

Table of Contents

- 1. Project Context and Objective**
- 2. Supervisor requirements analysis**
- 3. Automatic Allocation Logic**
- 4. Technical Architecture**
- 5. Sidebar Navigation Explanation**
- 6. Cargo management module**
- 7. Storage management module**
- 8. Digital Yard Module**
- 9. Manual Allocation Module**
- 10. AI Control Center**
- 11. Import Manifest system**
- 12. Operations Module**
- 13. AI Prediction Center**
- 14. Reports Module**
- 15. Future Development**
- 16. Conclusion**

1. Project Context and Objective

The internship project was carried out at Djen Djen Port – Jijel as part of the development of an intelligent terminal management solution.

The main objective was to design and implement a smart port storage optimization platform capable of assisting operators in managing cargo storage operations efficiently.

The developed system aims to improve the traditional storage workflow by providing intelligent assistance through:

- Cargo registration and lifecycle management.
- Storage zone and position management.
- Automatic allocation recommendation based on operational constraints.
- Terminal activity monitoring and decision support.
- Operational reporting and data visualization.
- AI-oriented modules for future prediction and optimization.

The platform provides a foundation for transforming manual storage management into a more organized, automated, and data-driven process

2. Supervisor Requirements Analysis

The project requirements describe a port divided into several operational zones.

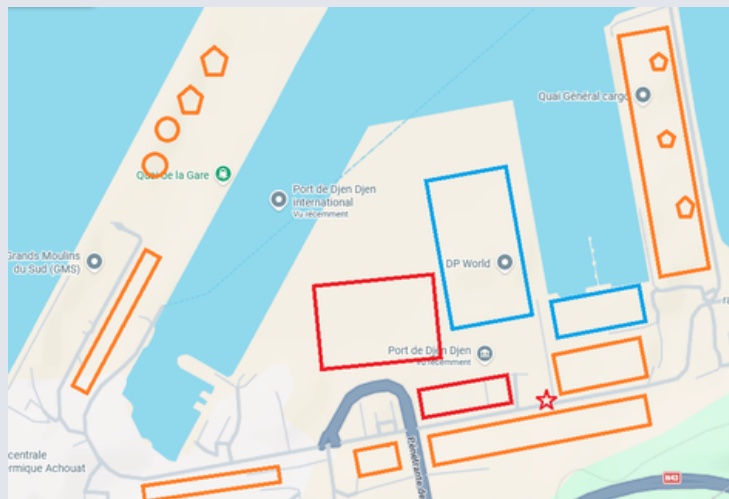
Cargo restrictions:

- Vehicles and containers must pass through a scanner before retrieval.
- Bulk cargo must be stored in silos.
- Packages can be stacked.
- Coils cannot be stacked.

The system must select compatible storage locations according to cargo properties and available capacity.

The attached terminal map defines fictional zones:

- Red areas: vehicle related zones.
- Blue areas: container zones.
- Orange areas: miscellaneous cargo zones.
- Scanner represented by a specific point.



3. Automatic Allocation Logic

The automatic allocation module is the core optimization component of the platform. Its objective is to automatically determine the most suitable storage position for each cargo unit while respecting terminal constraints and operational requirements.

When a cargo requires storage, the optimization engine analyzes all available positions and evaluates their compatibility with the cargo characteristics.

Evaluation Criteria :

The system considers several factors:

- **Cargo Compatibility:** Checks whether the storage location is suitable for the cargo type, category, handling requirements, and storage conditions.
- **Zone Restrictions:** Ensures that the cargo is assigned only to authorized zones according to terminal rules.
- **Position Availability:** Verifies that the storage position is free and has enough capacity.
- **Weight Limitations:** Compares cargo weight with the maximum supported capacity of the selected position.
- **Storage Rules:** Applies operational constraints such as stacking rules, cargo-specific restrictions, and accessibility requirements.

After evaluating all possible candidates, the system selects the position with the best compatibility score and returns it as an AI-based storage recommendation.

The selected allocation is then recorded in the system to maintain traceability, allowing operators to review decisions and analyze previous storage operations.

4. Technical Architecture

The platform is based on a three-layer architecture that separates the user interface, application logic, and data management.

Frontend Layer

Technology: React + Vite , Responsible for:

- Providing the user interface for terminal operators.
- Managing user interactions and actions.
- Displaying cargo, storage, AI, and reporting information.
- Communicating with the backend through API requests.

Backend Layer

Technology: FastAPI , Responsible for:

- Handling the application business logic.
- Managing database operations.
- Executing cargo allocation and optimization algorithms.
- Processing imported manifest files.
- Generating operational reports.

Database Layer , Responsible for storing and managing:

- Cargo information.
- Storage positions and zones.
- Allocation decisions.
- Movement history.
- Operational data.

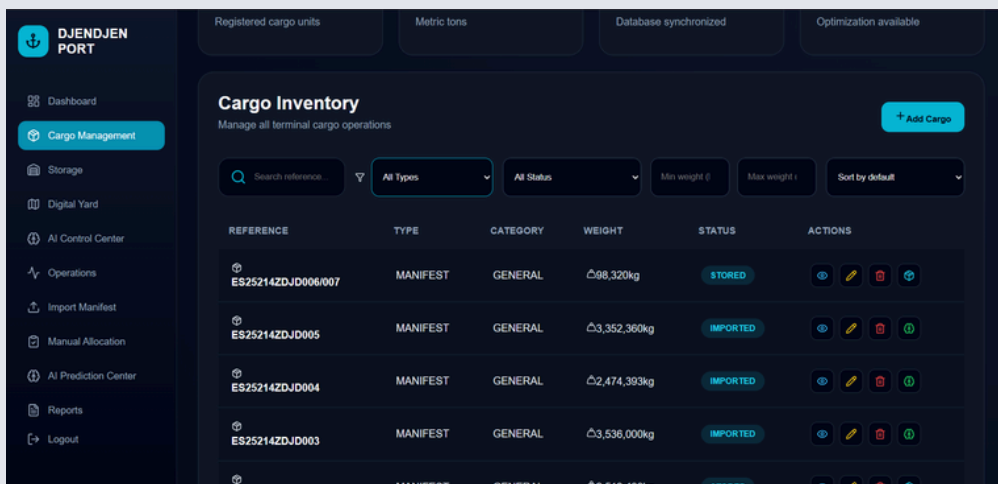
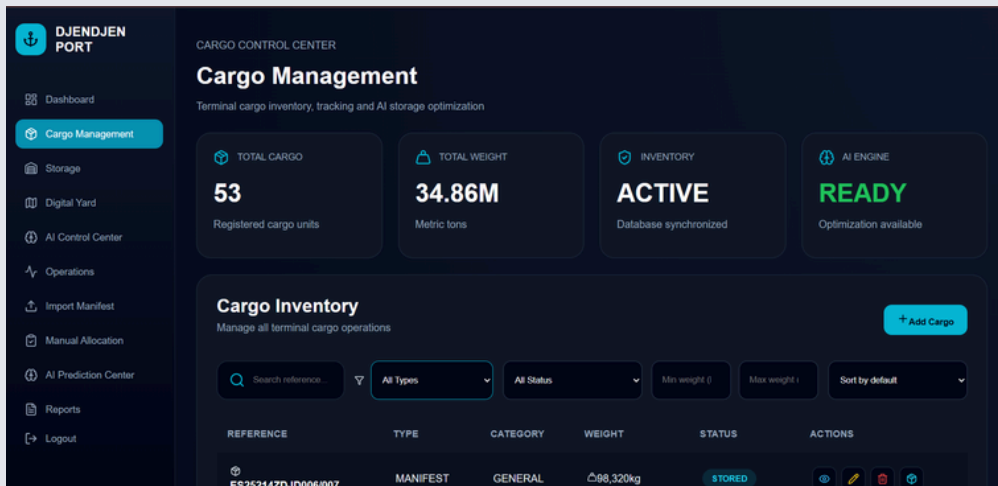
5. Sidebar Navigation Explanation

The sidebar provides quick access to all main modules of the platform. Each section corresponds to a specific operational function.

- **Dashboard :**Provides a global overview of the terminal status through key operational indicators and summarized information.
- **Cargo Management :** The central module for managing the complete cargo lifecycle, from registration and tracking to storage and release operations.
- **Storage :** Manages storage organization by displaying zones, positions, availability, and cargo placement information.
- **Digital Yard :** Provides a visual representation of the terminal layout and storage zones to facilitate operational understanding.
- **AI Control Center :** A monitoring interface that displays intelligent recommendations, allocation analysis, optimization decisions, and operational risk indicators.
- **Operations :** Tracks and monitors terminal activities, including cargo movements and operational actions.
- **Import Manifest :** Handles Excel manifest processing by importing cargo information and converting it into structured system data.
- **Manual Allocation :** Allows operators to manually assign cargo to storage positions when human intervention is required.
- **AI Prediction Center :** Provides a dedicated interface prepared for future predictive models such as storage forecasting and operational prediction.
- **Reports :** Generates operational reports and exports related to inventory, storage, movements, and AI decisions for analysis and traceability.

6. Cargo Management module

The Cargo Management page is the main module responsible for managing the complete lifecycle of cargo units inside the terminal. It allows operators to visualize, modify, analyze, and control cargo operations from a single interface.



Displayed Information:

For each cargo unit, the system displays:

Reference: Unique identifier of the cargo , **Type:** Nature and classification of the cargo , **Category:** Cargo group used for operational management , **Weight:** Physical weight information used for storage compatibility , **Status:** Current lifecycle state (Imported, Stored, etc.) , **Actions:** Available operations depending on cargo status.

Cargo Actions:

View Button : Displays complete cargo details through an information interface.

The operator can consult: Cargo identification information ,Physical characteristics ,Weight and dimensions ,Current status ,Storage and allocation information.

This allows better tracking and decision-making before performing any operation.

Edit Button : Allows authorized modification of cargo information. The updated data is validated and sent to the backend through API requests to ensure database consistency.

Delete Button : Removes incorrect or unnecessary cargo records after confirmation.

AI Button (Imported Cargo)

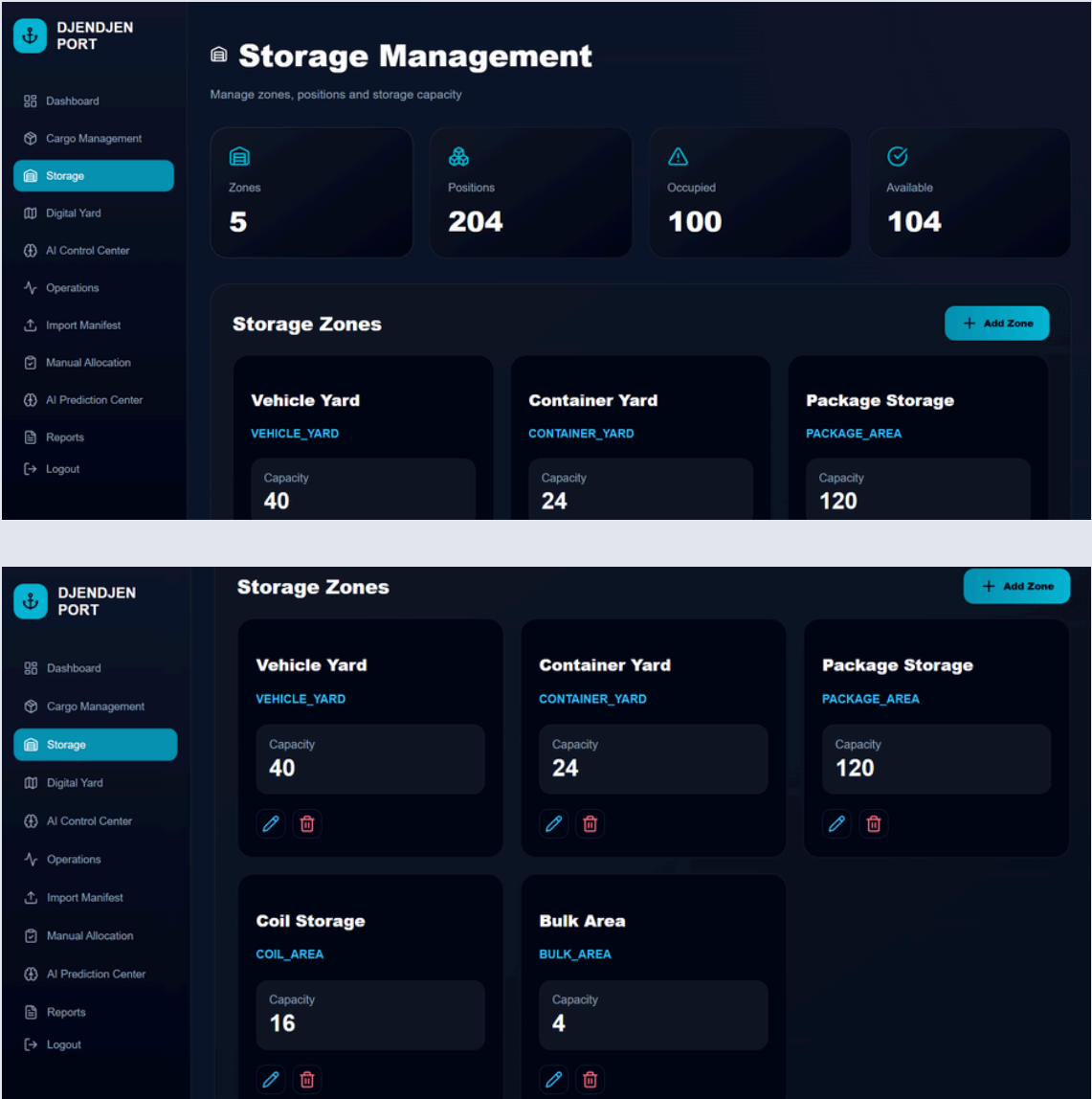
Available for cargo that has not yet been stored ,When activated the system requests an automatic storage recommendation by analyzing: Cargo characteristics ,Storage compatibility ,Available positions ,Operational constraints.The result is a recommended storage location to assist the operator's decision.

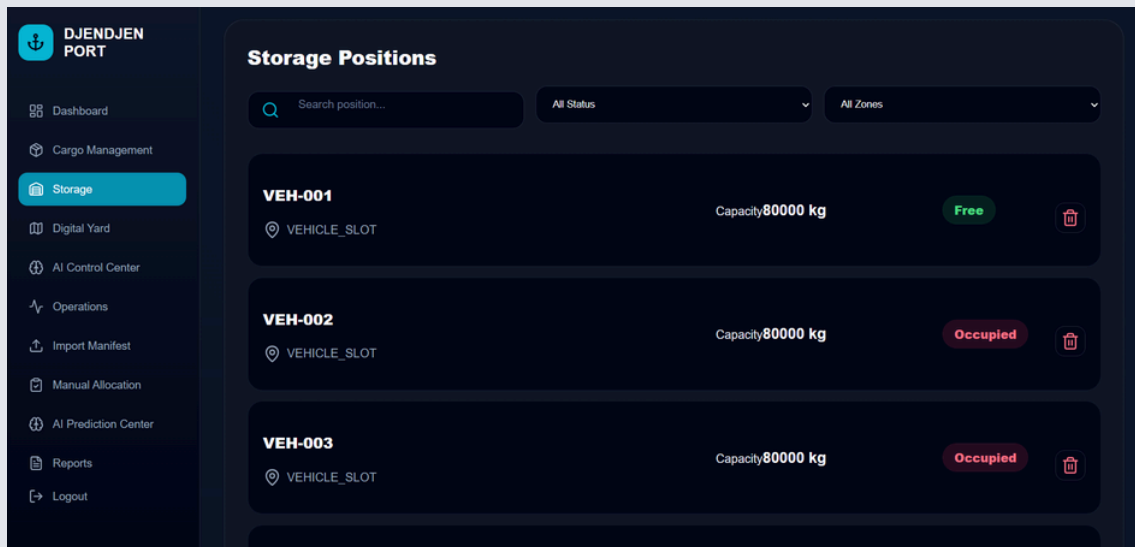
Release Button (Stored Cargo)

Available for cargo already stored in the terminal , It starts the release workflow by: Updating cargo status ,Recording the operational movement, Maintaining traceability of terminal activities. This ensures that every cargo movement is tracked inside the system.

7. Storage management module

The Storage module is responsible for managing and visualizing the terminal storage organization. It provides operators with a clear view of available storage resources and cargo placement.





The module manages:

Storage Zones:

Represents the different areas of the terminal dedicated to specific cargo operations.

Storage Positions:

Defines the individual locations where cargo can be placed.

Availability:

Displays free positions and available capacity for new cargo allocation.

Occupancy:

Shows currently occupied locations and stored cargo information.

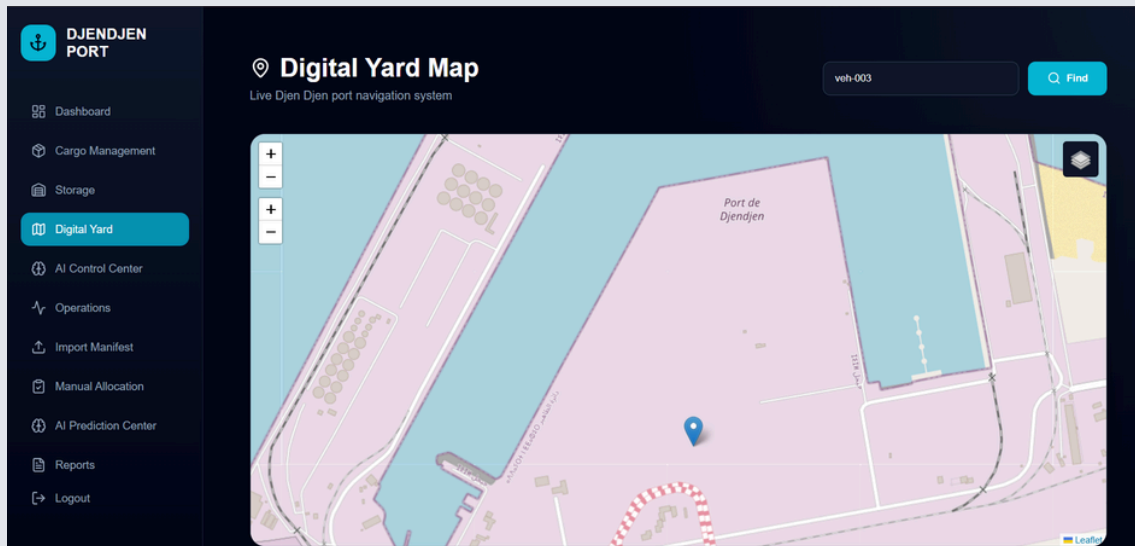
The storage information is continuously used by the allocation engine to identify compatible positions according to cargo characteristics, storage constraints, and available capacity.

This module provides the foundation for automatic allocation by ensuring that the system always has an updated view of the terminal storage state.

8. Digital Yard Module

The Digital Yard module provides a visual representation of the terminal storage environment. It allows operators to understand the physical organization of the yard and monitor storage areas in an intuitive way.

The objective of this module is to create a digital view of the terminal layout, helping operators quickly identify zones, positions, and storage distribution.



Main Functions :

Terminal Zone Visualization : Displays the different storage areas and their organization inside the yard.

Storage Position Representation : Provides a visual view of available and occupied positions.

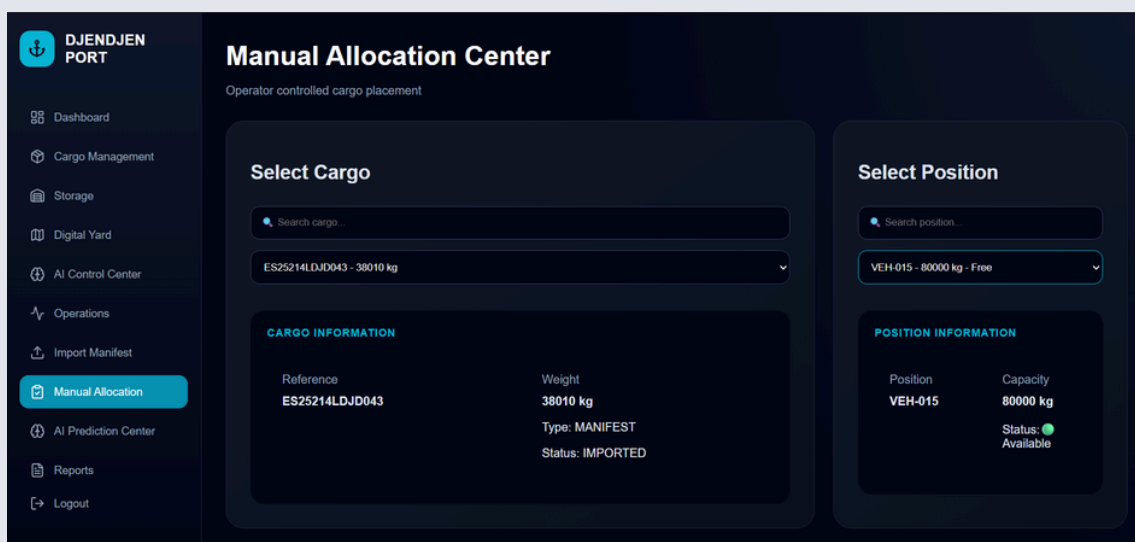
Operational Understanding : Helps operators understand cargo distribution and storage organization without relying only on numerical data.

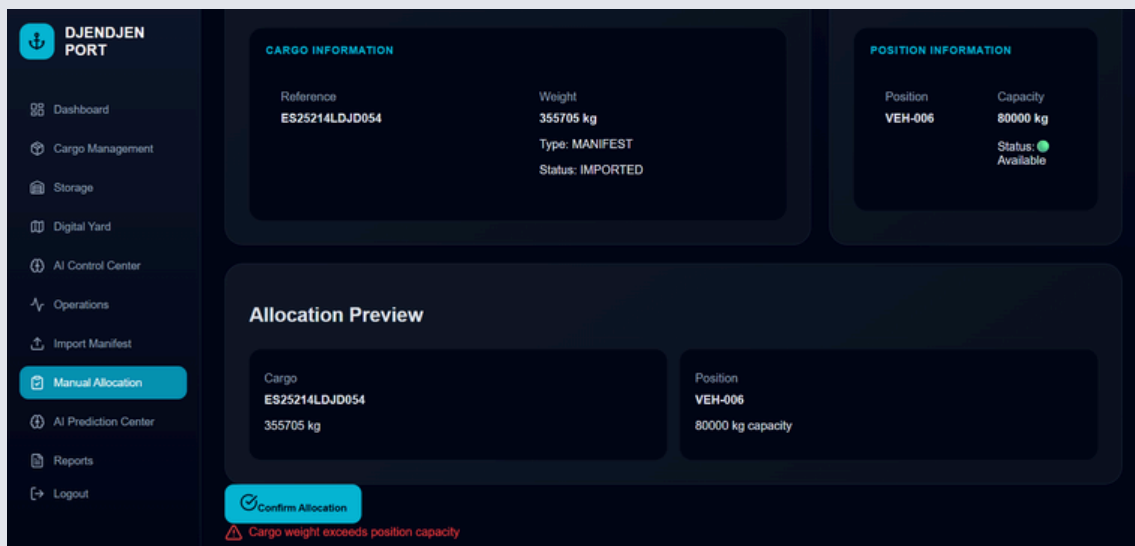
Storage Monitoring Support : The digital representation can be used with allocation and storage modules to improve operational decision-making.

9. Manual Allocation Module

The Manual Allocation module provides operators with the possibility to manually assign cargo to a specific storage position.

This module is designed to keep human control over storage operations when automatic allocation is not suitable or when an operator needs to apply a specific decision.





Main Objectives :

Human Control: Allows operators to manually choose a storage location based on operational requirements.

Operational Correction: Provides a solution for correcting or adjusting automatic allocation decisions when necessary.

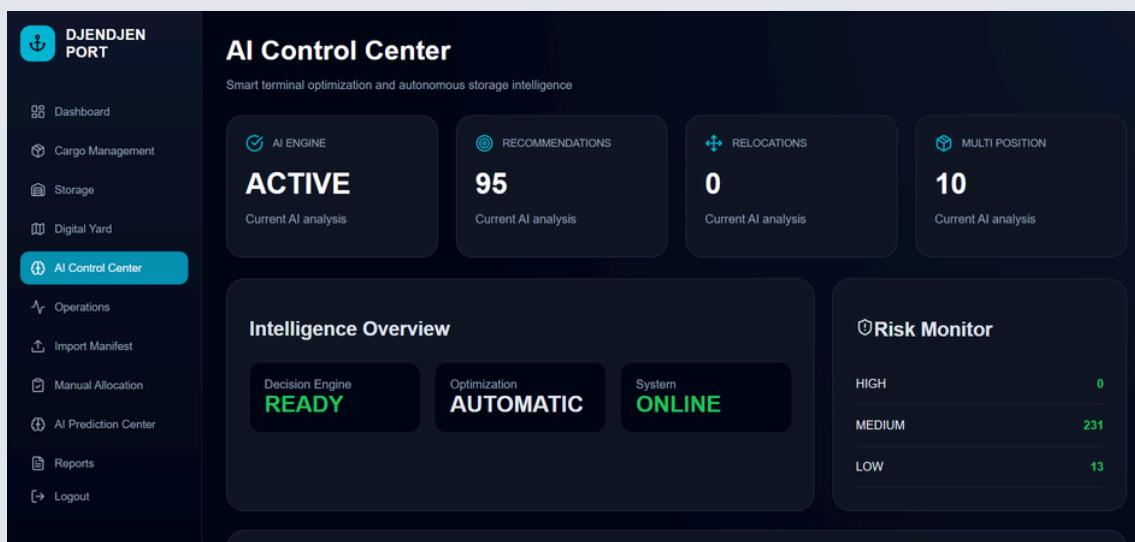
Manual Decision Recording: Each manual allocation is recorded in the system to maintain traceability of operational actions.

This module complements the AI allocation system by combining automatic optimization with operator expertise and real-world terminal requirements.

10. AI Control Center

The AI Control Center is the intelligent monitoring dashboard of the platform. It provides operators with a global view of AI-assisted operations, optimization decisions, and operational risk analysis.

This module transforms stored operational data into useful information to support decision-making and improve terminal efficiency.



DJENDJEN PORT - Dashboard - Cargo Management - Storage - Digital Yard - AI Control Center - Operations - Import Manifest - Manual Allocation - AI Prediction Center - Reports - Logout	Recent AI Decisions					
	CARGO REFERENCE	TYPE / CATEGORY	HANDLING	WEIGHT	POSITION / SCORE	CONFIDENCE
	ES25214ZDJ005 MANIFEST	GENERAL MANIFEST	STANDARD	100,000 kg	Position 65 AI Score: 125.6	MEDIUM
	ES25214ZDJ004 MANIFEST	GENERAL MANIFEST	STANDARD	100,000 kg	Position 65 AI Score: 125.81	MEDIUM
	ES25214ZDJ001 MANIFEST	GENERAL MANIFEST	STANDARD	76,660 kg	Position 185 AI Score: 166	HIGH
Pending AI Optimization Recommendations						
ES25214ZDJ005 MANIFEST - GENERAL Recommended Position #65 Weight 100,000 kg Handling STANDARD AI Confidence MEDIUM Why this position? ✓ Cargo requires storage allocation ✓ Weight capacity verified ✓ Handling method compatible ✓ Position optimized for retrieval AI Score: 125.6 AI_RECOMMENDED						

- Main Indicators :**
- The dashboard displays several key indicators:**
 - AI Engine Status:** Shows the current state of the intelligent system and confirms that AI analysis services are active.
 - Number of Recommendations:** Indicates the number of cargo allocation recommendations generated by the system.
 - Relocations:** Displays cargo movement and relocation activity to monitor operational changes.
 - Multi-position Operations:** Identifies cargo operations involving multiple storage positions, which may require additional attention.

- Risk Monitor :**
- The Risk Monitor provides an operational risk classification based on rule-based scoring.**
 - The current analysis considers several factors:**
 - Cargo State:** Evaluates the current lifecycle status of cargo.
 - Allocation Status:** Checks whether cargo allocation decisions are available and appropriate.
 - Storage Constraints:** Considers storage conditions and compatibility limitations.
 - Movement Activity:** Analyzes operational movements and relocation history.
 - The system classifies situations into different risk levels to help operators identify cargo requiring attention.**

- Future Evolution**
- The current architecture is designed to support future integration of machine learning models.**
 - Future improvements could include:** Predictive risk detection , Storage demand forecasting , Automatic anomaly detection , Learning from historical terminal operations.
 - This makes the AI Control Center a scalable module that can evolve from rule-based intelligence toward advanced predictive analytics.**

11. Import Manifest System

The Import Manifest system is responsible for transforming external Excel manifest files into structured cargo data that can be managed by the platform.

This module automates the transition from raw shipping information to organized terminal records.

Dashboard

Cargo Management

Storage

Digital Yard

AI Control Center

Operations

Import Manifest

Manual Allocation

AI Prediction Center

Reports

Logout

DATA IMPORT

Import Manifest

Upload, analyze and import vessel cargo manifests.

Upload Excel Manifest

Supported formats: XLSX / XLS

Choose File

testing_file.xlsx

testing_file.xlsx

Analyze Manifest

Manifest Analysis Completed

Dashboard

Cargo Management

Storage

Digital Yard

AI Control Center

Operations

Import Manifest

Manual Allocation

AI Prediction Center

Reports

Logout

Manifest Analysis Completed

Upload ID: 919dd029-27c5-4d90-9d7d-f9a61a91e362

Manifest validated successfully

Manifest Summary

53

Bill Of Lading

93

Vehicles

0

Containers

0

Trailers

146

Total Records

Bill Of Lading

Vehicles

Containers

Trailers

Validation

BL Number	Cargo Nature	Description	Weight	Packages
ES25214LDJD001	8	CHARPENTE METALLIQUE: HEA, IPE	250432	
ES25214LDJD002	8	02 UNITS ZZ3256V364MC1 SINOTRUK SITRAK 6X4 DUMP TRUCK (VOIR BL)	28440	
ES25214LDJD003	8	01 UNIT 35CBM U-TYPE TIPPER SEMI TRAILER WITH THREE AXLES ZJV9700UZX (VOIR BL)	11300	
ES25214LDJD005	8	02 UNITS SEMI-TRAILER (VOIR BL)	30300	

Dashboard

Cargo Management

Storage

Digital Yard

AI Control Center

Operations

Import Manifest

Manual Allocation

AI Prediction Center

Reports

Logout

ES25214LDJD049	LZGJLJM40USA088792	B178	B178MU003	2025	12670
ES25214LDJD049	LZGJLJM42SX088793	B178	B178M0003	2025	12670
ES25214LDJD049	LZGJLJM44SX088794	B178	B178M0003	2025	12670
ES25214LDJD049	LZGJLJM46SX088795	B178	B178M0003	2025	12670
ES25214LDJD049	LZGJLJM48SX088796	B178	B178M0003	2025	12670
ES25214LDJD060	LZGJD8J18SX095246	B178	B178M0093	2025	4770
ES25214LDJD060	LEFYFCC24SHN37783	B109	B109M0022	2025	4770
ES25214LDJD060	LEFYECG36SHN32509	B109	B109M0067	2025	4770

Confirm Database Import

Import Completed

Status: IMPORT_COMPLETED

File: testing_file.xlsx

Backend Modules :

The import process is divided into several specialized modules:

excel reader : Responsible for reading and extracting data from uploaded Excel files.

analyzer : Analyzes the structure of the spreadsheet and identifies available information.

column mapper : Maps Excel columns to the corresponding system fields to ensure correct data integration.

validator : Checks the imported information and verifies data consistency before insertion.

database importer : Inserts validated cargo records into the database and creates the required system entities

Import Workflow

1. Excel Manifest Upload : The operator uploads the shipping manifest file containing cargo information.

2. Spreadsheet Reading : The system reads the Excel file and extracts its content.

3. Column Analysis : The platform analyzes the file structure to identify available columns and their meaning.

4. Data Mapping : Excel fields are matched with internal database fields such as: Cargo reference, Type, Category, Weight, Quantity, Other operational information.

5. Data Validation : The system verifies that the imported information is complete and compatible with the platform requirements.

6. Database Insertion : After successful validation, cargo records are automatically created and become available in the Cargo Management module.

Vehicle Information Handling :

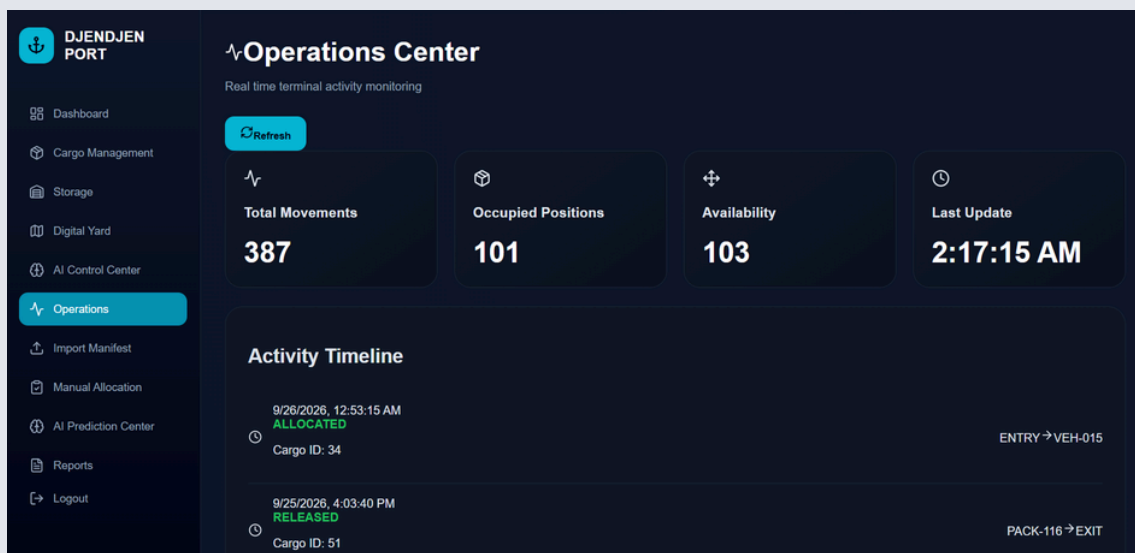
When the cargo nature corresponds to vehicles, the system applies a specific processing workflow to store vehicle-related information separately.

This allows the platform to manage different cargo categories while respecting their operational characteristics and storage requirements.

12. Operations Module

The Operations module is responsible for monitoring and managing terminal activities. It provides visibility over cargo movements and operational actions performed inside the storage system.

The objective of this module is to maintain operational traceability by recording and displaying the different actions carried out during the cargo lifecycle.



Main Functions:

Movement Monitoring : Displays cargo movements and operational changes between storage locations.

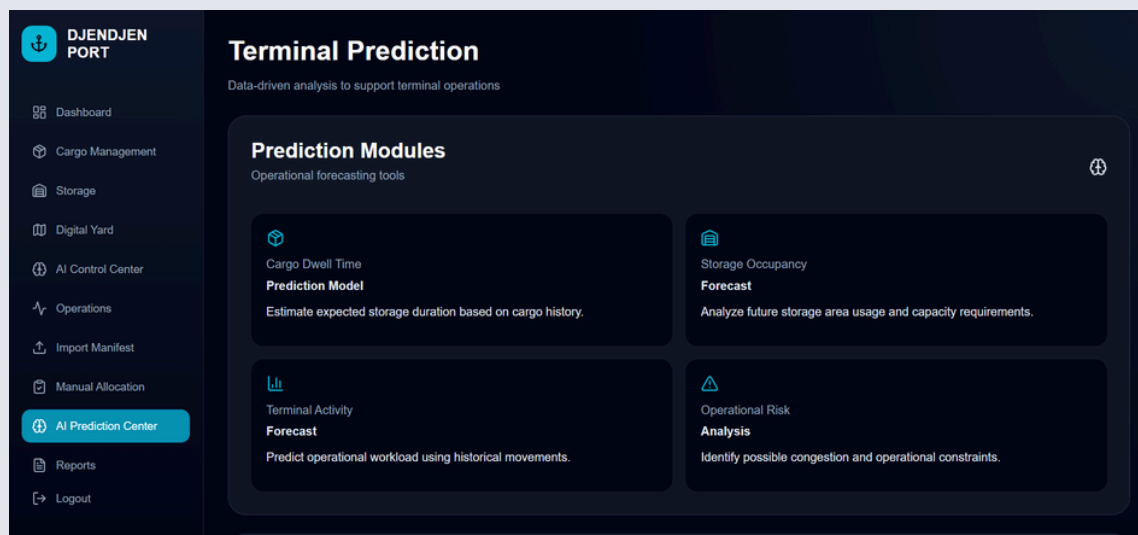
Activity Tracking : Keeps a history of performed operations to ensure that every action is recorded and traceable.

Operational Status Monitoring : Provides information about current activities and helps operators follow terminal workflow.

Decision Traceability : Links operational actions with allocation decisions and storage activities.

13. AI Prediction Center

The AI Prediction Center is a dedicated interface designed to prepare the integration of future predictive intelligence modules into the terminal management system. Its objective is to use historical operational data to anticipate future events, improve planning, and support decision-making.



Included Predictive Modules:

- **Cargo Dwell Time Prediction** : Predicts the expected duration of cargo storage based on previous operational patterns.
- **Storage Occupancy Forecast** : Estimates future storage utilization to help anticipate capacity issues and improve space management.
- **Terminal Activity Forecast** : Analyzes operational trends to predict future cargo movements and workload.
- **Operational Risk Prediction** : Aims to detect potential operational problems by analyzing historical activities and performance indicators.

Current Implementation Status :

The current version provides the user interface and the required integration structure for future AI prediction models.

Note:

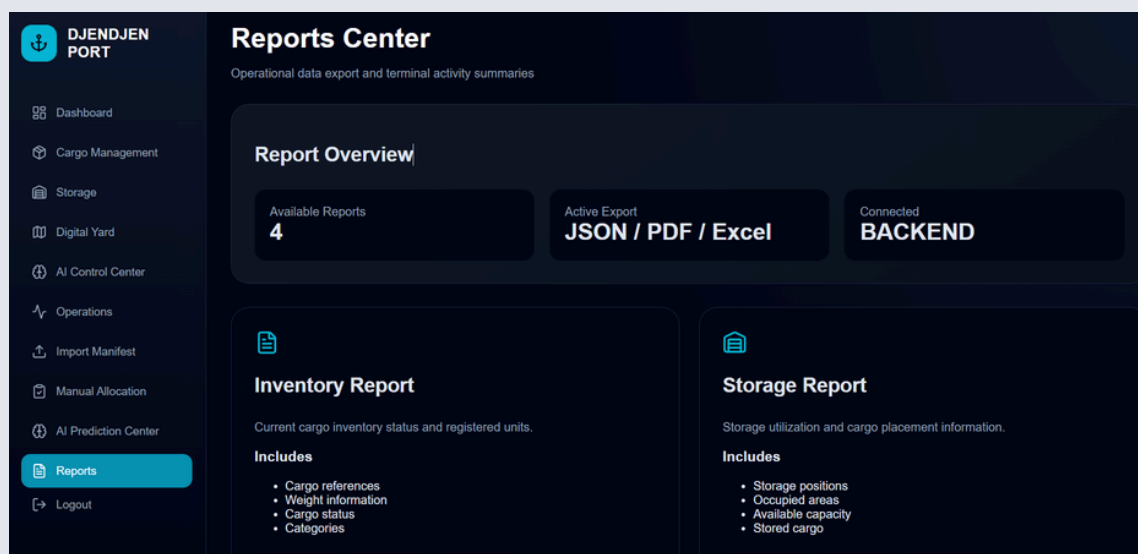
The predictive models are not yet implemented because the system does not currently have sufficient historical operational data from the company. Machine learning models require a large amount of real historical data (cargo movements, storage duration, occupancy evolution, operational events, etc.) to be trained and produce reliable predictions.

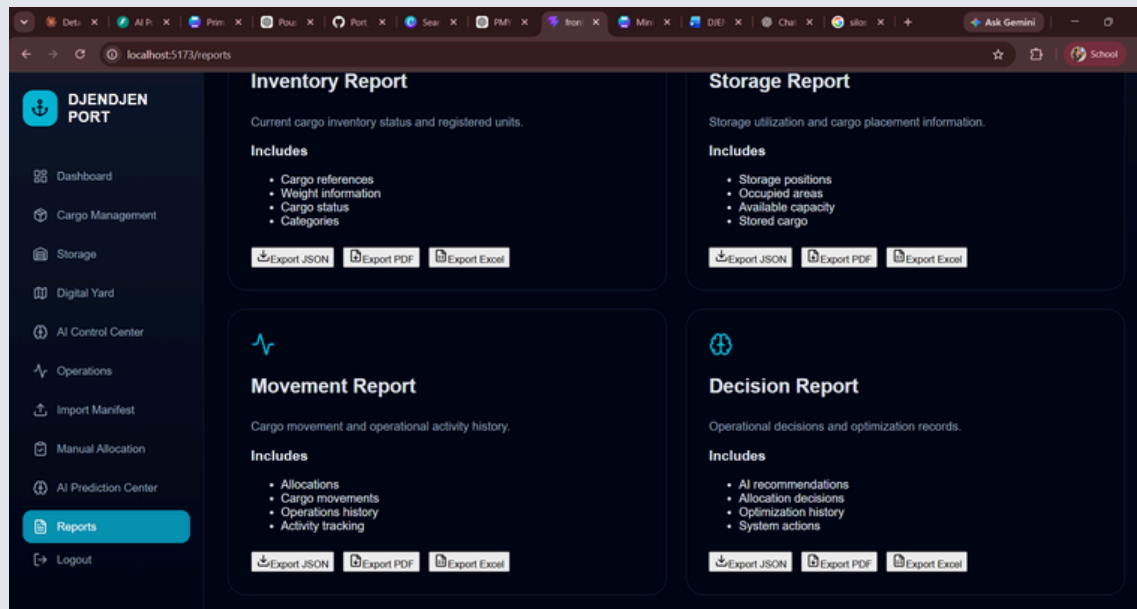
Once enough historical data is collected, this module can be extended by integrating trained predictive models connected to the existing platform.

14. Reports Module

The Reports module is responsible for generating operational reports from the data stored in the terminal management system. It provides operators with a centralized interface to extract, analyze, and archive important operational information.

The objective of this module is to improve traceability, monitoring, and decision support by transforming system data into structured reports.





Available Reports:

Inventory Report : Provides a complete overview of cargo currently managed by the system, including: Cargo references ,Cargo types and categories ,Weight information ,Current status

Storage Report : Provides information about storage organization and capacity: Storage positions, Occupancy status, Available capacity, Cargo placement information.

Movement Report : Displays the history of cargo operations: Cargo movements, Position changes, Operational actions, Activity history.

Decision Report : Provides information about intelligent decisions generated by the system: AI recommendations, Allocation decisions, Optimization results.

Export System : The module supports multiple export formats:

JSON Export: Provides structured data that can be reused by other applications or systems.

PDF Export: Generates professional documents suitable for reporting and operational documentation.

Excel Export: Allows further analysis and manipulation of data using spreadsheet tools.

15. Future Development

Although the current platform provides intelligent storage management and decision support, several improvements can be integrated in future versions to transform it into a more advanced smart port system

Possible Improvements :

- **Real Historical Learning Models :** Integration of machine learning models trained on real port operational data to improve allocation decisions and predictions.
- **Predictive Storage Duration :** Prediction of how long cargo will remain in storage to optimize space planning and retrieval operations.
- **Congestion Forecasting :** Anticipation of storage saturation and operational bottlenecks before they occur.
- **Real-Time IoT Integration :** Connection with sensors and terminal equipment to collect live information about cargo movements and storage conditions.
- **Advanced Anomaly Detection :** Automatic detection of unusual situations such as abnormal delays, excessive movements, or operational conflicts.
- **Digital Twin of Terminal Operations :** Creation of a virtual representation of the terminal to simulate operations, analyze scenarios, and improve decision-making.

16. Conclusion

The developed AI Port Storage Optimization Platform provides a solid foundation for intelligent terminal management by combining cargo management, automatic allocation assistance, AI-based monitoring, risk analysis, and operational reporting within a scalable architecture.

However, the development process was carried out with a limited amount of operational data, as only a single manifest file was available during the internship period. This limitation affected the quantity of available information for analysis and reduced the possibility of implementing certain advanced features, particularly those requiring historical data such as predictive models, machine learning algorithms, and more accurate operational forecasting.

Despite these limitations, the project successfully established the necessary architecture and intelligent decision-support mechanisms required for future improvements.

With access to larger and more complete historical datasets from the port, the platform can be further enhanced by integrating advanced prediction models, improving allocation optimization, and developing more accurate AI-based operational analysis.

The objective is therefore to continue evolving this solution toward a complete Smart Port management system capable of supporting more efficient, predictive, and data-driven terminal operations.